Rotations

1 P

2 P

3 D

4 M

5 P

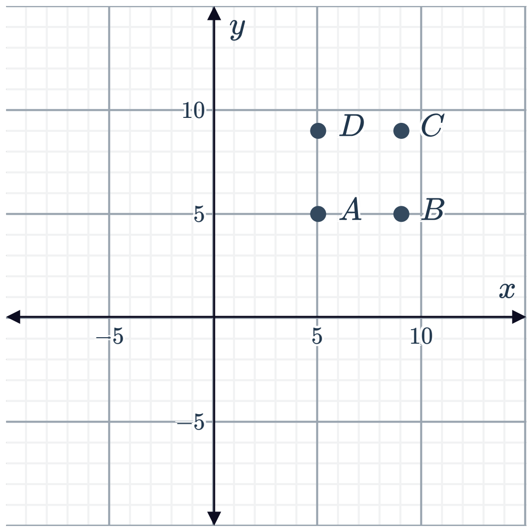
6 G

7 $(0, -2)$

8

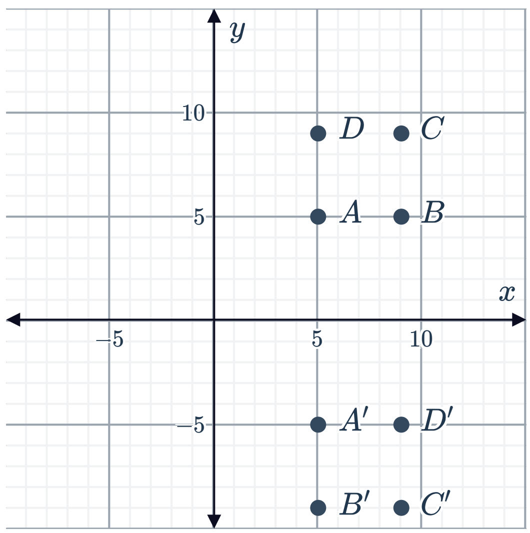
a

i



y
x
10
5
−5
10
−5
D
C
A
B

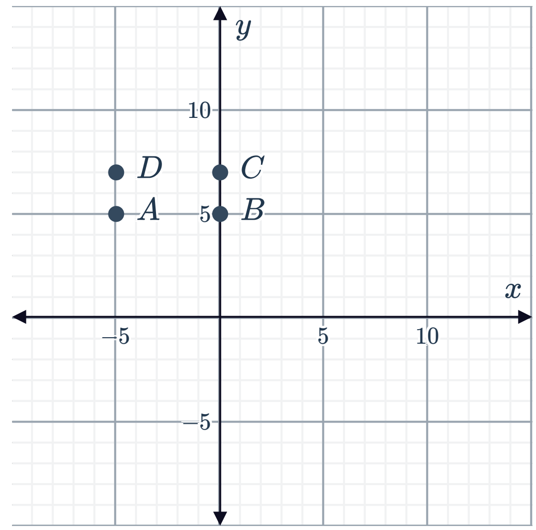
ii



y
x
10
5
−5
10
−5
D
C
A
B
A′
D′
B′
C′

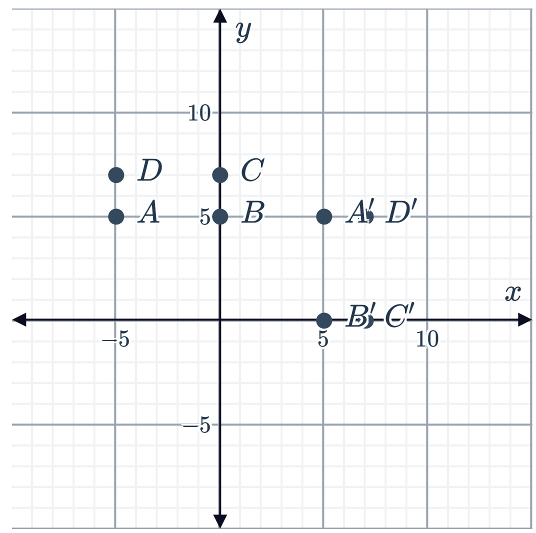
b

i



y
x
10
5
−5
10
−5
D
C
A
B

ii

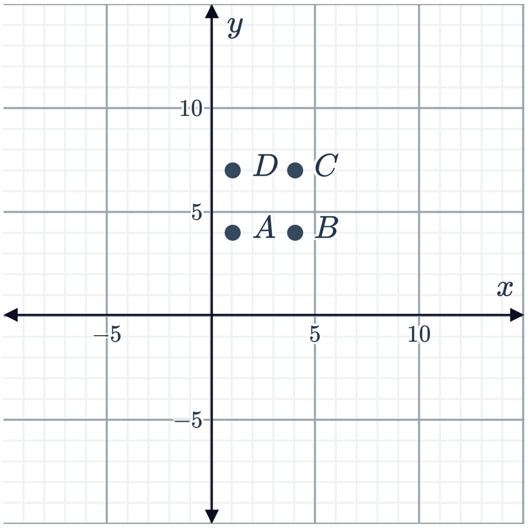


y
x
10
5
−5
10
−5
D
C
A
B
A′
D′
B′
C′

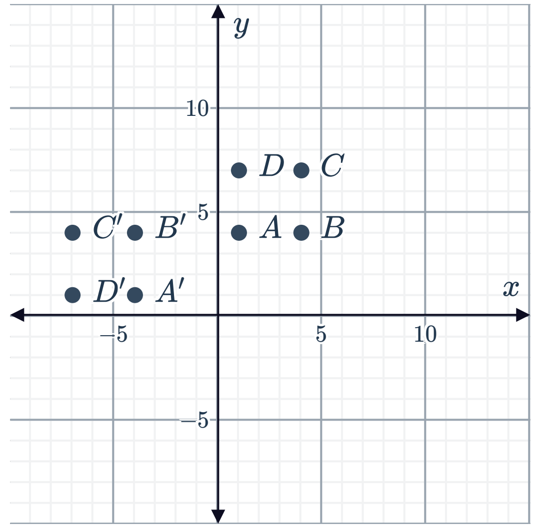
9

a

i

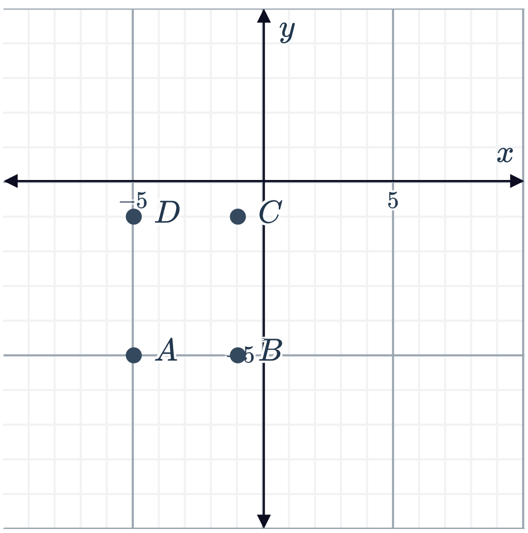

y
x
10
5
−5
−5
5
10
D
C
A
B

ii

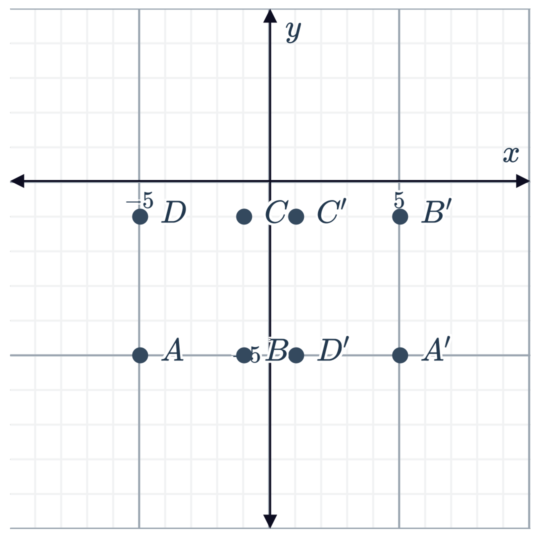

y
x
10
5
−5
−5
5
10
D
C
A
B
C′
B′
D′
A′

b

i

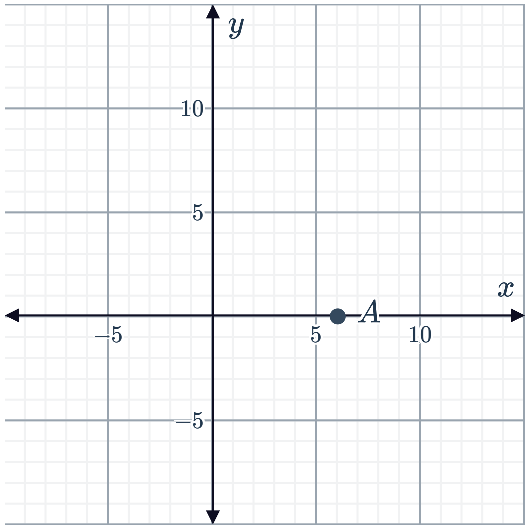

y
x
−5
5
−5
D
C
A
B

ii


y
x
−5
5
−5
D
C
C′
B′
A
B
D′
A′

10

a



b $B\,(0, -6)$

c 6-unit translation horizontally to the left
6-unit translation vertically downwards

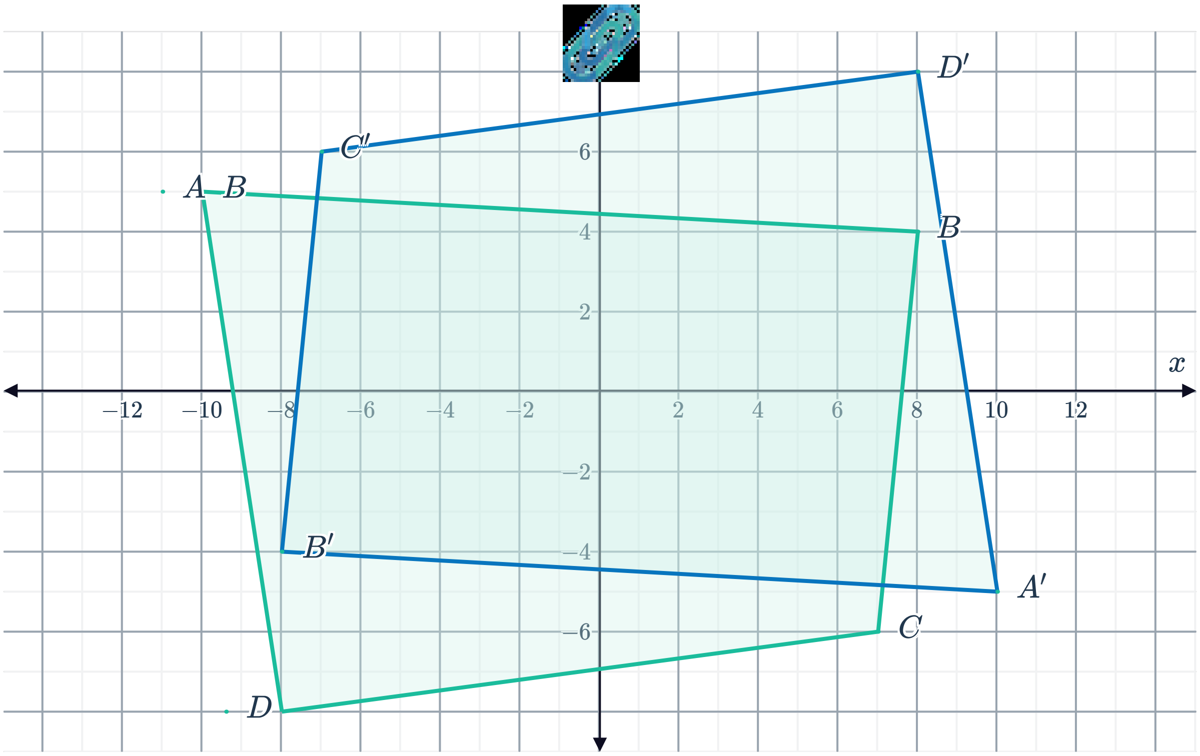
Additional questions

11 The shape itself does not change, has been rotated one quadrant clockwise. It is the equivalent of turning the graph on its right side and relabelling the x and y axes.

12 $\triangle P$

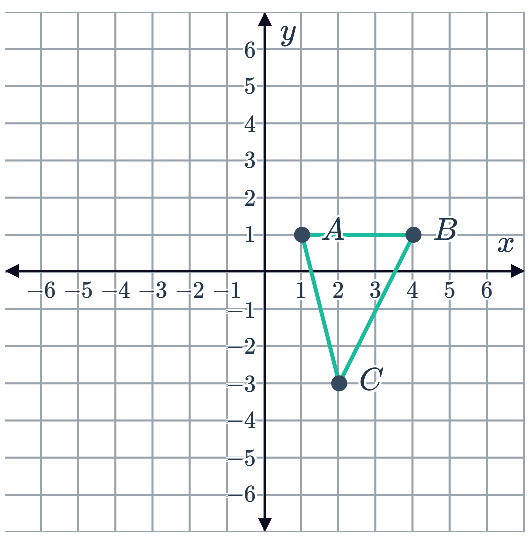
13 $(4, -4)$

14

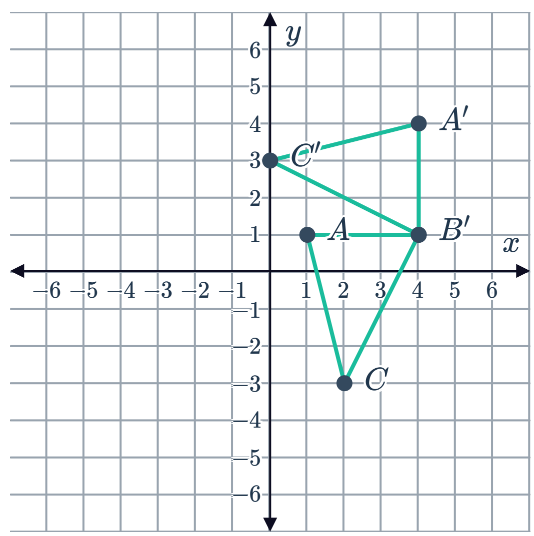


15

a



b



16 Figure A was rotated 90° clockwise about the origin.

17

a Yes, when rotated 180° each face is showing the dots in the same arrangement as before they were rotated.

b No, consider the face with one/four/five dot(s). Rotating that 90 gives the exact same face.

18 Avery needs to turn the map 180° clockwise or counterclockwise. He is currently facing

19 Yes, Aneta rotated the segment clockwise around the origin instead of counterclockwise.